In [ ]:
```
EXPERIMENT 7 – HANDLING OUTLIERS & MISSING VALUES IN A DATASET
PIN NO: 241UDS0109
AIM:-
To preprocess a dataset by removing duplicates, detecting outliers, and handling missing values using
Pandas and Scikit-learn SimpleImputer
```

In [ ]:
```
## Step 1: Import the Basic Libraries
```

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:
```
## Step 2: Load & Analyze the Dataset
```

In [2]:
```
df_241UDS0109 = pd.read_csv("Experiment_7_241UDS0109_Student_Profile.csv")
df_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 62 entries, 0 to 61
Data columns (total 24 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   Branch                 60 non-null     object 
 1   Marks[10th]            60 non-null     float64
 2   Marks[12th]            60 non-null     float64
 3   Gender                 60 non-null     object 
 4   Board[10th]            60 non-null     object 
 5   Board[12th]            60 non-null     object 
 6   Category               60 non-null     object 
 7   GPA 1                  60 non-null     float64
 8   Rank                   61 non-null     float64
 9   Normalized Rank        60 non-null     float64
 10  CGPA                   60 non-null     float64
 11  Current Back           60 non-null     float64
 12  Ever Back              59 non-null     float64
 13  GPA 2                  60 non-null     float64
 14  GPA 3                  60 non-null     float64
 15  GPA 4                  60 non-null     float64
 16  GPA 5                  60 non-null     float64
 17  GPA 6                  60 non-null     float64
 18  Olympiads Qualified    60 non-null     float64
 19  Technical Projects     60 non-null     float64
 20  Tech Quiz              60 non-null     float64
 21  Engg. Coaching         60 non-null     float64
 22  NTSE Scholarships      60 non-null     float64
 23  Miscellany Tech Events  60 non-null     float64
dtypes: float64(19), object(5)
memory usage: 11.8+ KB
```

In [ ]:
```
### Display the first five records
```

In [3]:
```
df_241UDS0109.head()
```

Out[3]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	EEE	84.65	86.99	Female	CBSE	CBSE	SC
1	IT	83.57	74.97	Female	State Board	ISC	GEN
2	NaN	82.63	82.92	Male	State Board	CBSE	EWS
3	ECE	83.24	87.05	Male	ICSE	ISC	GEN
4	EEE	52.00	80.95	Male	State Board	CBSE	OBC

Columns 9–15 of 22 · first column repeated

	GPA 1	Rank	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	7.85	64917.0	0.68334	...	8.45	7.56	7.81
1	8.05	50252.0	0.52897	...	7.79	7.85	7.95
2	7.20	2413.0	0.02540	...	7.31	7.08	7.55
3	7.39	16287.0	0.17144	...	8.13	8.03	7.67
4	8.05	59055.0	0.62163	...	7.97	7.99	8.46

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	7.92	0.0	5.0	0.0	0.0	1.0	3.0
1	7.87	3.0	4.0	3.0	1.0	0.0	3.0
2	7.06	0.0	3.0	0.0	1.0	1.0	1.0
3	7.62	1.0	5.0	3.0	0.0	1.0	0.0
4	7.96	2.0	3.0	3.0	1.0	1.0	3.0

5 rows × 24 columns

In [ ]:
```
### Check the shape
```

In [4]:
```
df_241UDS0109.shape
```

Out[4]: (62, 24)

In [ ]:
```
### Remove completely empty rows
```

In [5]:
```
df_241UDS0109 = df_241UDS0109.dropna(how="all")
df_241UDS0109.shape
```

Out[5]: (61, 24)

In [ ]:
```
## Step 3: Find & Remove the Duplicates
```

In [6]:
```
df_241UDS0109[["Marks[10th]", "Marks[12th]", "CGPA"]].duplicated().sum()
```

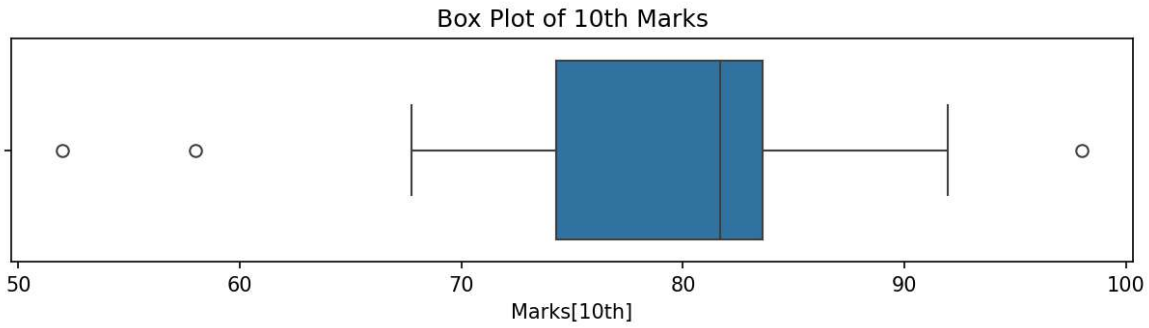
Out[6]: np.int64(1)

In [7]:
```
df_241UDS0109 = df_241UDS0109.drop_duplicates()
df_241UDS0109.shape
```

Out[7]: (60, 24)

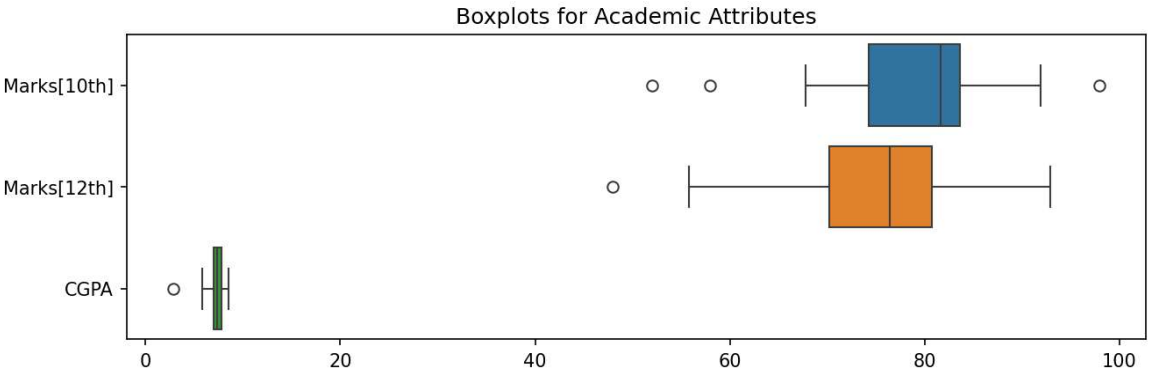
In [ ]:
```
## Step 4: Identify & Remove the Outliers
### Box plot for 10th marks
```

In [8]:
```
plt.figure(figsize=(10, 2), dpi=150)
sns.boxplot(data=df_241UDS0109, x="Marks[10th]")
plt.title("Box Plot of 10th Marks")
plt.show()
```



In [ ]:
```
### Create a list for outlier columns and visualize with boxplots
```

In [9]:
```
columns_241UDS0109 = ["Marks[10th]", "Marks[12th]", "CGPA"]
plt.figure(figsize=(10, 3), dpi=150)
sns.boxplot(data=df_241UDS0109[columns_241UDS0109], orient="h")
plt.title("Boxplots for Academic Attributes")
plt.show()
```



In [ ]:
```
### Calculate the IQR ranges Q1 and Q3
```

In [10]:
```
Q1_241UDS0109 = df_241UDS0109[columns_241UDS0109].quantile(0.25)
Q3_241UDS0109 = df_241UDS0109[columns_241UDS0109].quantile(0.75)
Q1_241UDS0109, Q3_241UDS0109
```

Out[10]:
```
(Marks[10th]    74.270
 Marks[12th]    70.235
 CGPA            6.965
 Name: 0.25, dtype: float64,
 Marks[10th]    83.605
 Marks[12th]    80.805
 CGPA            7.775
 Name: 0.75, dtype: float64)
```

In [ ]:
```
### Calculate IQR
```

In [11]:
```
IQR_241UDS0109 = Q3_241UDS0109 - Q1_241UDS0109
IQR_241UDS0109
```

Out[11]:
```
Marks[10th]     9.335
Marks[12th]    10.570
CGPA            0.810
dtype: float64
```

In [ ]:
```
### Find outliers using IQR range
```

In [12]:
```
outliers_241UDS0109 = (
    (df_241UDS0109[columns_241UDS0109] < (Q1_241UDS0109 - 1.5 * IQR_241UDS0109)) |
    (df_241UDS0109[columns_241UDS0109] > (Q3_241UDS0109 + 1.5 * IQR_241UDS0109))
)
outliers_241UDS0109.sum()
```

Out[12]:
```
Marks[10th]    3
Marks[12th]    1
CGPA           1
dtype: int64
```

In [ ]:
```
### Check shape before and after removing outliers
```

In [13]:
```
df_241UDS0109.shape
```

Out[13]:
```
(60, 24)
```

In [14]:
```
df_241UDS0109 = df_241UDS0109.loc[~outliers_241UDS0109.any(axis=1)].copy()
df_241UDS0109.shape
```

Out[14]:
```
(55, 24)
```

In [ ]:
```
## Step 5: Fill the Missing Values
### Find missing values
```

In [15]:
```
df_241UDS0109.isnull().sum()
```

Out[15]:
```
Branch                   1
Marks[10th]              1
Marks[12th]              1
Gender                   1
Board[10th]              1
Board[12th]              1
Category                 1
GPA 1                    1
Rank                     0
Normalized Rank          1
CGPA                     0
Current Back             1
Ever Back                2
GPA 2                    1
GPA 3                    1
GPA 4                    1
GPA 5                    1
GPA 6                    1
Olympiads Qualified      1
Technical Projects       1
Tech Quiz                1
Engg. Coaching           1
NTSE Scholarships        1
Miscellany Tech Events   1
dtype: int64
```

In [ ]:
```
### Fill Marks[10th] with mean value
```

In [16]:
```
df_241UDS0109["Marks[10th]"] = df_241UDS0109["Marks[10th]"].fillna(
    df_241UDS0109["Marks[10th]"].mean()
)
df_241UDS0109["Marks[10th]"].isnull().sum()
```

Out[16]:
```
np.int64(0)
```

In [ ]:
```
### Import SimpleImputer
```

In [17]:
```
from sklearn.impute import SimpleImputer
```

In [ ]:
```
### Apply median imputation to remaining numerical columns
```

In [18]:
```
numeric_columns_241UDS0109 = df_241UDS0109.select_dtypes(include="number").columns
numerical_imputer_241UDS0109 = SimpleImputer(strategy="median")
df_241UDS0109[numeric_columns_241UDS0109] = numerical_imputer_241UDS0109.fit_transform(
    df_241UDS0109[numeric_columns_241UDS0109]
)
df_241UDS0109[numeric_columns_241UDS0109].isnull().sum()
```

Out[18]:
```
Marks[10th]              0
Marks[12th]              0
GPA 1                    0
Rank                     0
Normalized Rank          0
CGPA                     0
Current Back             0
Ever Back                0
GPA 2                    0
GPA 3                    0
GPA 4                    0
GPA 5                    0
GPA 6                    0
Olympiads Qualified      0
Technical Projects       0
Tech Quiz                0
Engg. Coaching           0
NTSE Scholarships        0
Miscellany Tech Events   0
dtype: int64
```

In [ ]:
```
### Fill Branch with mode using SimpleImputer
```

In [19]:
```
df_241UDS0109["Branch"].value_counts()
```

Out[19]:
```
Branch
CSE      13
ECE      10
CIVIL     8
EEE       6
IT        6
MECH      6
PROD      5
Name: count, dtype: int64
```

In [20]:
```
categorical_imputer_241UDS0109 = SimpleImputer(strategy="most_frequent")
df_241UDS0109[["Branch"]] = categorical_imputer_241UDS0109.fit_transform(df_241UDS0109[["Branch"]])
df_241UDS0109["Branch"].value_counts()
```

Out[20]:
```
Branch
CSE      14
ECE      10
CIVIL     8
EEE       6
IT        6
MECH      6
PROD      5
Name: count, dtype: int64
```

In [ ]:
```
### Fill remaining categorical missing values
```

In [21]:
```
categorical_columns_241UDS0109 = df_241UDS0109.select_dtypes(include=["object"]).columns
categorical_imputer_all_241UDS0109 = SimpleImputer(strategy="most_frequent")
df_241UDS0109[categorical_columns_241UDS0109] = categorical_imputer_all_241UDS0109.fit_transform(
    df_241UDS0109[categorical_columns_241UDS0109]
)
```

In [ ]:
```
### Check all remaining missing values
```

In [22]:
```
df_241UDS0109.isnull().sum()
```

Out[22]:
```
Branch                  0
Marks[10th]             0
Marks[12th]             0
Gender                  0
Board[10th]             0
Board[12th]             0
Category                0
GPA 1                   0
Rank                    0
Normalized Rank         0
CGPA                    0
Current Back            0
Ever Back               0
GPA 2                   0
GPA 3                   0
GPA 4                   0
GPA 5                   0
GPA 6                   0
Olympiads Qualified     0
Technical Projects      0
Tech Quiz               0
Engg. Coaching          0
NTSE Scholarships       0
Miscellany Tech Events  0
dtype: int64
```

In [ ]:
```
### Final cleaned dataset information
```

In [23]:
```
df_241UDS0109.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 55 entries, 0 to 59
Data columns (total 24 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Branch                  55 non-null     object
 1   Marks[10th]             55 non-null     float64
 2   Marks[12th]             55 non-null     float64
 3   Gender                  55 non-null     object
 4   Board[10th]             55 non-null     object
 5   Board[12th]             55 non-null     object
 6   Category                55 non-null     object
 7   GPA 1                   55 non-null     float64
 8   Rank                    55 non-null     float64
 9   Normalized Rank         55 non-null     float64
 10  CGPA                    55 non-null     float64
 11  Current Back            55 non-null     float64
 12  Ever Back               55 non-null     float64
 13  GPA 2                   55 non-null     float64
 14  GPA 3                   55 non-null     float64
 15  GPA 4                   55 non-null     float64
 16  GPA 5                   55 non-null     float64
 17  GPA 6                   55 non-null     float64
 18  Olympiads Qualified     55 non-null     float64
 19  Technical Projects      55 non-null     float64
 20  Tech Quiz               55 non-null     float64
 21  Engg. Coaching          55 non-null     float64
 22  NTSE Scholarships       55 non-null     float64
 23  Miscellany Tech Events  55 non-null     float64
dtypes: float64(19), object(5)
memory usage: 10.7+ KB
```

In [ ]:
```
RESULT:
Duplicates were removed, outliers were identified using the IQR method and removed, and missing numerical
and categorical values were imputed using mean, median, and most-frequent strategies.
```